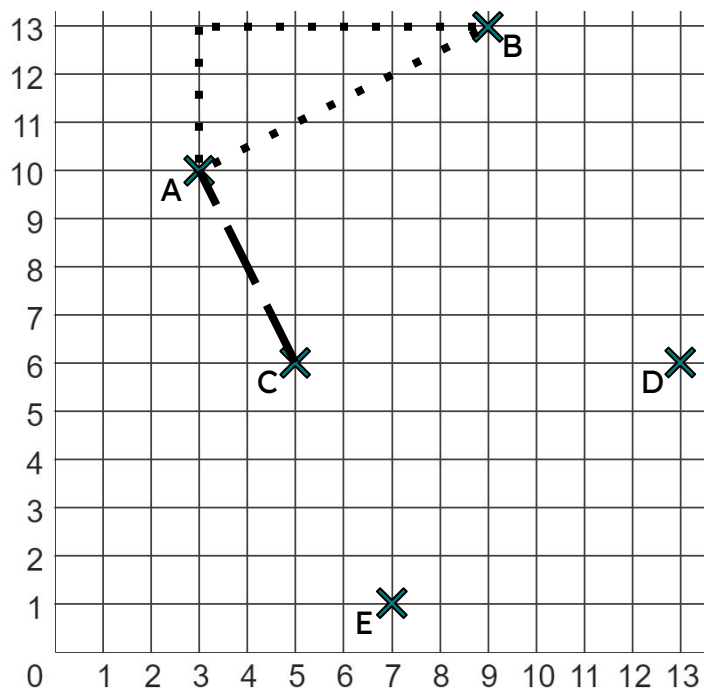




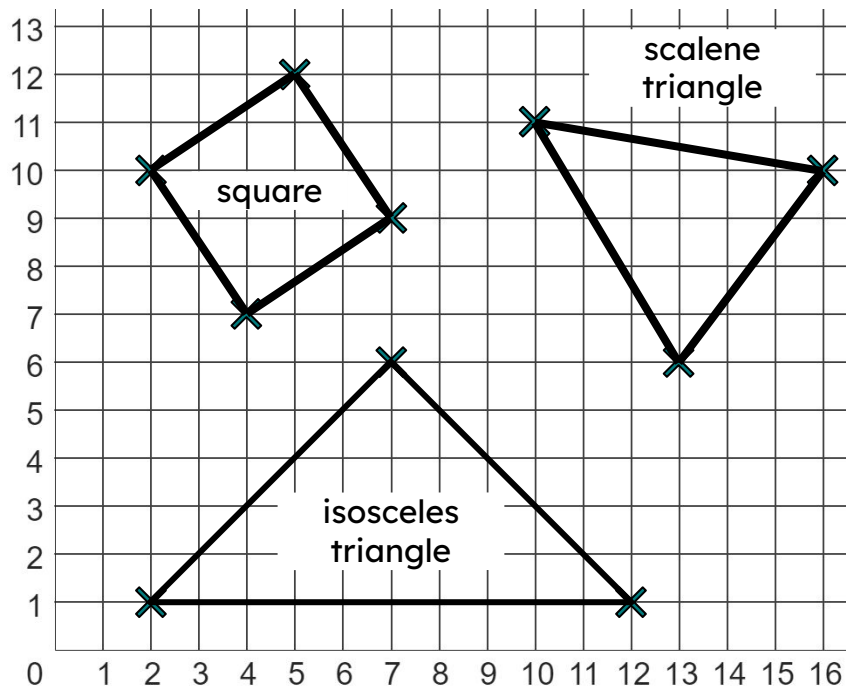
Problem solving with similarity and Pythagoras' theorem

Task A: Pythagoras' theorem with coordinates



1) Use Pythagoras' theorem to find the shortest distance (to 1 d.p) between points:

- a) A and B *6.7 units*
- b) A and C *4.5 units*
- c) B and D *8.1 units*
- d) C and D *8 units*
- e) B and E *12.2 units*



2) Calculate the perimeter of each shape (to 2 d.p.).

square
 4×3.60555
14.42 units

isosceles triangle
 $2 \times 7.071 + 10$
24.14 units

scalene triangle
 $5.83095 + 5 + 6.08276$
16.91 units

3) Write down three coordinates whose vertices make a triangle with a perimeter of between 20 and 22 units in length.
 Show this is correct using Pythagoras' theorem.

There are many possible solutions, such as a triangle with vertices (4, 6), (3, 12) and (10, 8)



Task B: Pythagoras' theorem with ratios

1) Each of these three widescreen TVs have width and height in the ratio 16 : 9.

a) Complete each sentence by choosing the appropriate word below.

Each of the TVs is similar to the others.

This is because their width and height are in proportion to each other.

The scale factor from screen A to B is 1.5.

b) By using Pythagoras' theorem, complete the ratio table for screen B (3 d.p.)

width	height	diagonal
16	9	18.358
10.459	5.883	12

c) Hence, calculate the width of screens A and C (2 d.p.)

screen A: $10.459 \div 1.5 = 6.97$ inches

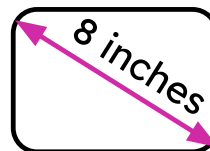
screen C: $10.459 \times 2.5 = 26.15$ inches

2) Izzy wants to buy this ultra widescreen TV, which has a width and height in the ratio 22 : 9.

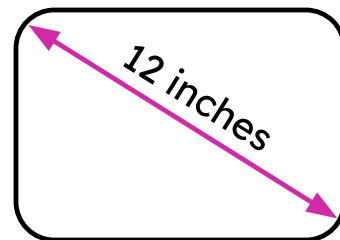
The TV has a 35 inch screen, and a stand that is 2 inches high.

Will the TV fit inside her TV cabinet?

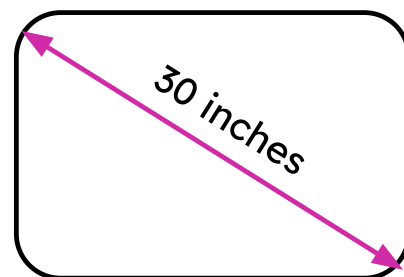
width	height	diagonal
22	9	23.77
32.39	13.25	35



screen A

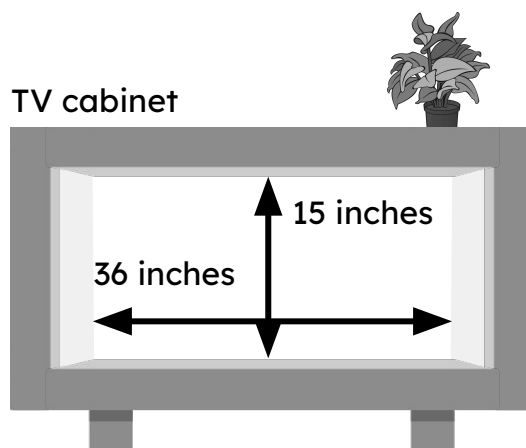


screen B



screen C

TV cabinet



No, the TV will not fit in the cabinet. Whilst the screen is less than the width and height of the cabinet, the TV's 2 inch stand makes the TV too tall.

3) Jacob sees three TVs for sale, each having a 28 inch screen.

retro TV (4 : 3 aspect ratio)

widescreen TV (16 : 9 aspect ratio)

ultra widescreen TV (22 : 9 aspect ratio)

Jacob wants to buy the TV whose screen has the largest area.
Which TV should he buy?

summary table of solutions

<i>TV</i>	<i>width</i>	<i>height</i>	<i>area</i>
<i>retro</i>	<i>22.4</i>	<i>16.8</i>	<i>376 inches²</i>
<i>widescreen</i>	<i>24.4</i>	<i>13.7</i>	<i>335 inches²</i>
<i>ultra widescreen</i>	<i>25.9</i>	<i>10.6</i>	<i>275 inches²</i>



Even though all three TVs have a diagonal length of 28 inches, the retro TV has the screen with the largest area.

